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FACULDADE DE
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Generative AI
2025/2026 - 2nd Semester

Project 2: Image-to-Prompt

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1 Introduction

We present here the second practical project, part of the students' evaluation process of the Generative AI course of the University of Coimbra. This work is to be done autonomously by a group of **two students**. The quality of your work will be judged as a function of the value of the technical work, the written description, and the public defence.

All sources used to perform the work, including code, pretrained components, external repositories, datasets, and **Generative AI tools**, must be clearly identified. If generative AI systems, e.g., code assistants, large language models, or image understanding models, are used during development, their use must be explicitly acknowledged and described. Students remain fully responsible for the correctness, originality, and integrity of all submitted material.

The **deadline** for delivering the work is the **8th of June 2026**.

The document may be written in Portuguese or in English and the report must follow the **Springer LNCS (Lecture Notes in Computer Science) format**. The official LaTeX template can be obtained from the Springer website[1], and an online version is also available on Overleaf[2].

The **written report** is limited to **12 pages** using the LNCS template excluding references. The report should be well structured and include the following elements: a general **introduction**, a **description of the problem**, the **methodology and approach**, the **experimental setup**, an **analysis of the results**, and a **conclusion**. The final mark is given individually.

Plagiarism will not be allowed and, if detected, it implies failing the course. While doing the work and when submitting it, you should pay particular attention to:

- clear description of the adopted approach and all external components;
- reproducible experiment protocol, including model versions, hyperparameters, and random seeds;
- quantitative evaluation using image-side similarity metrics;
- qualitative analysis of successes, failures, and ambiguous cases.

[1] https://www.springer.com/gp/computer-science/lncs/conference-proceedings-guidelines

[2] https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwpvhwnvfj

2 Problem Statement

In this project, the goal is **image-to-prompt inversion**. Students are given a small test set of images produced by the Latent Consistency Model `SimianLuo/LCM_Dreamshaper_v7`[14], a fast latent diffusion generator derived from DreamShaper / Stable Diffusion-style models[21].

For each target image, the generation setup is fixed and known: model checkpoint, scheduler family, number of inference steps, guidance scale, resolution, and random seed. The unknown variable is the text prompt. The task is to recover prompts that, when rendered again with the same generator and configuration, produce images visually close to the targets.

This is not a classification task and not an open-ended image captioning task. The recovered prompt must be judged by the image it produces when re-rendered by the LCM generator.

3 Objective

The main objective is to produce, for each target image, a ranked list of **three candidate prompts**. Each prompt must be rendered with the fixed LCM setup, producing three reconstructed images per target.

- Use the target images provided in the folder `lcm_results/tp2-chosen`.
- For each target image, return the **top-3 prompts** found by your system.
- For each returned prompt, provide the **LCM-rendered image** produced from that prompt.
- Rank the three candidates according to an image-side similarity criterion.
- Compare results using a fixed evaluation protocol across all target images.

Different approaches to this problem exist, including vision-language captioning, prompt sampling, prompt retrieval, LLM-based prompt refinement, gradient-based discrete prompt optimisation, and soft-prompt inversion. Students may choose their own approach or combine approaches, provided that the final submission returns discrete text prompts. The reference section provides a sample list of approaches related to the task for this practical assignment. Moreover, refer to theoretical slide packs: T6 to T9.

The work should **focus on a single final approach**. The report should present the method that the group considers to be its best solution, together with the corresponding results and evaluation. It is not necessary to document multiple alternative approaches, failed experiments, or extensive intermediate trials. The emphasis should be on clearly explaining the chosen approach, justifying its main design decisions, and reporting the final prompts, generated images, and metric values obtained with that approach.

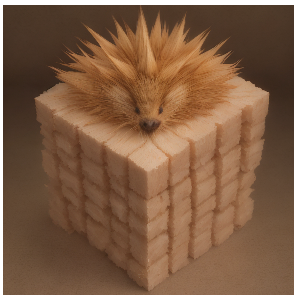
3.1 Target Set

The test set is composed of the six target images shown in Figure 1. The original prompts are not provided to students.

1159_25.png

1159_29.png

1159_3.png



1159_7.png

7836.png

9338.png

Figura 1: Target images for Project 2.

3.2 Generator Setup

All candidate prompts must be evaluated by rendering with the same image generator:

- Model: `SimianLuo/LCM_Dreamshaper_v7`
- Resolution: 768×768 **RGB**

- Random seed: encoded in the image filename. For example, `7836.png` uses seed **7836**, and `1159_25.png` uses seed **1159**.
- Guidance scale: **8.0**
- LCM origin steps: **50**
- Inference steps: **8**

The generator must be treated as deterministic under this setup. If the same prompt is rendered twice with the same configuration, the same image should be obtained up to implementation-level numerical differences.

3.3 Evaluation Metrics and Protocol

Evaluation is performed by comparing each target image with the image rendered from each recovered prompt. Metrics must be computed at the **image level**; prompt fluency alone is not sufficient. The mandatory quantitative metrics are:

- **CLIP image-image similarity** using a fixed CLIP image encoder[20] (e.g. openai/clip-vit-large-patch14) ;
- **LPIPS** perceptual distance (using Alexnet)[32];
- **Pixel RMSE** pixel level similarity;

For each target image, report:

- the target image;
- the three recovered prompts, ordered from best to worst;
- the three rendered images;
- a per-candidate metric table.

Across the full test set, report the mean and standard deviation of the main metrics. If stochastic components are used by the prompt search procedure, repeat the procedure with multiple optimiser seeds while keeping the LCM rendering seed fixed.

4 Deliverables

The assignment must be submitted as a single `.zip` file with the following filename format:

`IAGTP2_<student-number-1>_<student-number-2>.zip`

For example, for a group with students 12345 and 67890, the submitted file should be named:

`IAGTP2_12345_67890.zip`

The submitted `.zip` file must include:

- the report in `PDF` format;
- the source code used to generate, evaluate, and select the submitted prompts;
- a `README` file with clear instructions for running the code, including dependencies, main commands, and the expected folder structure.

Submitting the code is mandatory. The report should describe the proposed approach clearly, but it does not replace the code submission. Submissions without code may be penalized or considered incomplete. The submission deadline is:

8th of June 2026

5 Conclusion

This project emphasizes prompt inversion as an optimisation and evaluation problem in generative AI. The central challenge is not to describe an image in natural language, but to find prompts that reproduce target images under a fixed text-to-image generator. The final discussion should clearly connect the adopted method, the rendered reconstructions, and the measured image-side similarity.

Good luck!

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